

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B.Tech

Branch : All Branches

Semester : I

Subject Code & Name: Engineering Graphics (24AF1EGES104)

Max Marks: 60

Date:23/7/2025

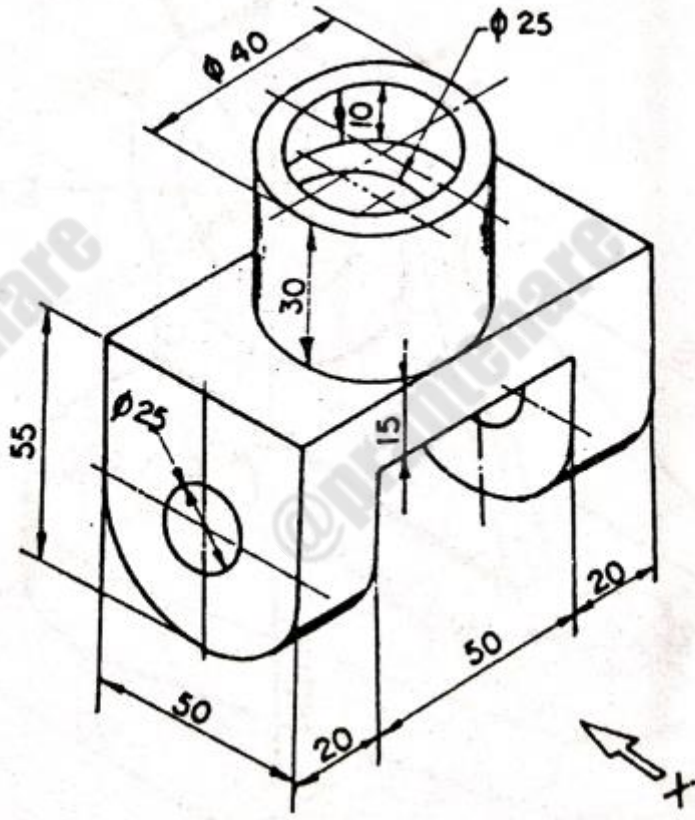
Duration: 4 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions.		12
1	What is the primary principle of orthographic projection? a) Parallel projection lines b) Converging projection lines c) Random projection lines d) Curved projection lines	CO5	1
2	What is the purpose of diagonal scale? a) To measure time accurately b) To measure sub units precisely c) To draw orthogonal views d) To measure angles	CO2	1
3	Representative fraction is the ----- a) ratio of the length in drawing to the actual length b) ratio of the actual length to the length in drawing c) reciprocal of actual length d) square of the length in drawing	CO2	1
4	A circle is placed at 20 degrees with vertical the view from top view will be ---- a) line b) Circle c) Ellipse d) Oval	CO5	1
5	Which of the following is a solid of revolution? a) Prism b) Pyramid c) Tetrahedron d) Cylinder	CO4	1
6	To understand some of the hidden geometry of components an imaginary plane is used to cut the object which is called ----- a) auxiliary plane b) picture plane c) section plane d) additional plane	CO4	1
7	A line inclined to both planes appears as a) True length b) Foreshortened length c) Full length d) None of the above	CO3	1

8	When the projectors are parallel to each other and also perpendicular to the plane, the projection is called _____.				CO5	1
	a) perspective projection	b) oblique projection	c) isometric projection	d) orthographic projection		
9	1st angle projection the _____ lies between _____ and _____.				CO5	1
	a) object, projection plane, observer	b) projection plane, object, observer	c) reference line, side view, front view	d) observer, object, plane of projection		
10	Which method of development is employed in case of cones?				CO6	1
	a) Parallel-line development	b) Approximation method	c) Triangulation development	d) Radial-line development		
11	What is the standard ratio of length to height for a closed, filled-in arrowhead in engineering drawings?				CO1	1
	1:3	3:1	2:1	1:1		
12	In a first angle projection, the top view is located:				CO5	1
	(a) Above the front view	(b) Below the front view	(c) To the left of the front view	(d) To the right of the front view		
Q. 2	Solve the following.					12
A)	Construct a diagonal scale of R.F. is 1/4000 to show meters and long enough to measure up to 500 meters.				CO2	6
B)	Differentiate Aligned and Uni-directional system of placing the dimensions on a drawing with the help of diagrams.				CO1	6
Q. 3	Solve the following.					12
A)	A line AB has its end A, 10 mm above the HP and 24 mm in front of the VP. End B is 55 mm above the HP and 50 mm in front of the VP. The distance between the end projectors is 45 mm. Draw the projections of the line and find the TL and true inclinations. Also locate the traces.				CO3	6
B)	Draw the projections of the following points on the same ground line. (i) Point A, 20 mm above HP, 25 mm behind VP; (ii) Point B, 25 mm below HP, 20 mm behind VP (iii) Point C, 20 mm below HP, 30 mm in front of VP (iv) Point D, 20 mm above HP, 25 mm in front of VP (v) Point E, on HP, 25 mm behind VP (vi) Point F, on VP, 30 mm above HP.				CO3	6
Q.4	Solve any one the following.					12
A)	Draw the projection of a regular hexagon of 25 mm side having one of its side in the HP and inclined at 60° to the VP and its surface making an angle of 45° with the HP .				CO3	12
B)	A square prism, base 40 mm side and height 65 mm, has its axis inclined at 45° to the HP and has an edge of its base, on the HP and inclined at 30° to the VP. Draw its Projections.				CO4	12

Q.5	Solve any one the following.		12
A)	A hexagonal pyramid, base 30 mm side and axis 65 mm long, is resting on its base on the H.P. with two edges parallel to the V.P. It is cut by a section plane, perpendicular to the V.P. inclined at 45° to the H.P and intersecting the axis at a point 25 mm above the base. Draw the front view, sectional top view, sectional side view and true shape of the section.	CO5	12
B)	A right circular cone, diameter of base 60 mm and axis height is 60 mm is resting on its base on HP .it is cut by a section plane perpendicular to VP and inclined to HP at 45° and bisecting the axis of the cone. Draw the front view, top view and development of surfaces.	CO6	12
Q. 6	Solve Any One of the following.		12
A)	Figure 1 shows pictorial view of an object. Draw following views by using Third angle method of projection: i) Front view looking in the direction Xii) Top View iii) Side view from left	CO5	12
 Fig.1.			
B)	Draw Isometric view & show overall dimensions.	CO5	12

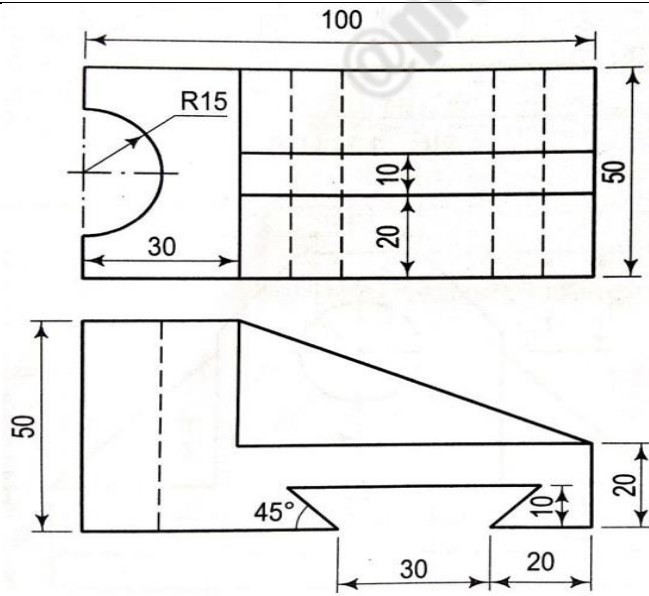


Fig.2.

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